PS-003 | Federated traffic-flow prediction

Federated Traffic Intelligence

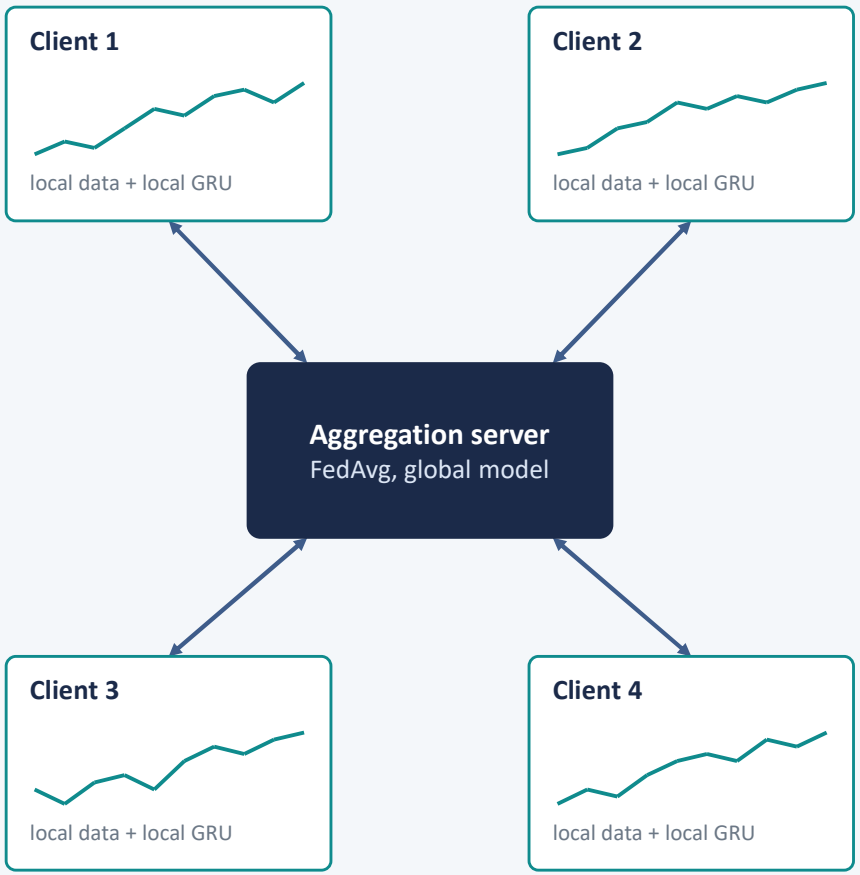
AI-powered traffic prediction through collaborative learning

TEAM NAME: Team NoCivicSense

MEMBERS: Bareeq Qayoom, Stacy Anna Dsouza, Jyothi Mandala

PROBLEM: PS-003

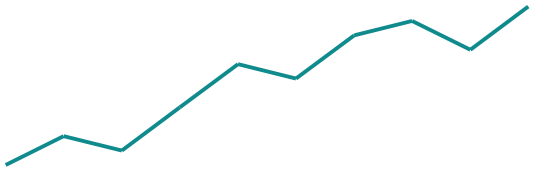
DOMAIN: AI/ML, Federated Learning, Networking, Smart Cities



01 THE PROBLEM

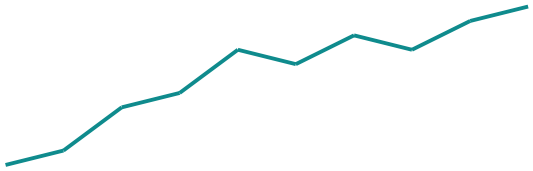
Traffic data is split across many owners

City traffic agency



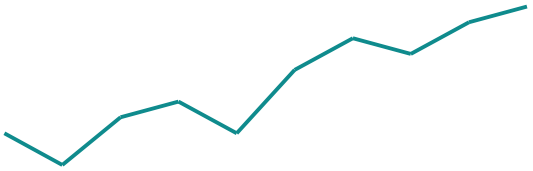
keeps its own traffic records

Highway operator



keeps its own traffic records

Private data company



keeps its own traffic records

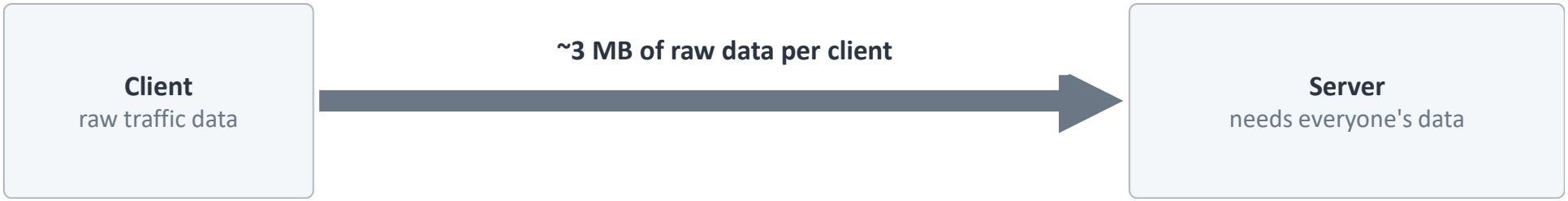
Pooling it all in one place is costly, slow, and often not allowed.

So, can we still build one good traffic predictor?

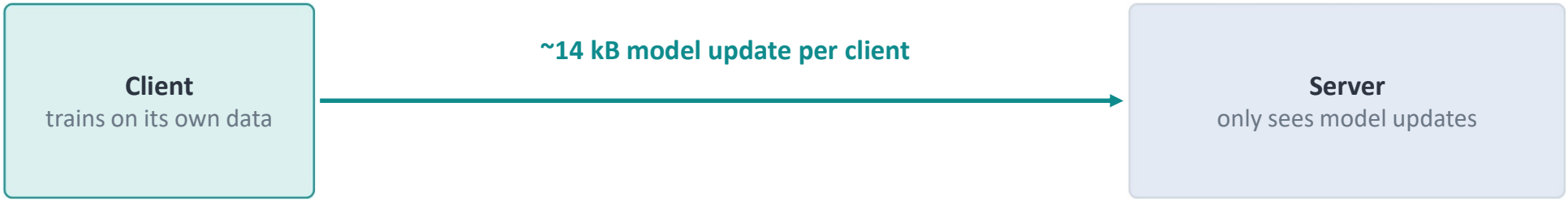
02 OUR IDEA

Share the learning, not the data

USUAL WAY: SEND THE RAW DATA



OUR WAY: SEND ONLY THE MODEL UPDATE



The algorithm: FedAvg plus a GRU

THE FORECASTER

GRU

A small neural network that reads traffic readings in order and predicts what comes next.

+

THE TEAM-TRAINER

FedAvg

Federated Averaging: each site trains the model on its own data, and the server averages the results into one model.

Together: a traffic forecaster trained by many sites, without sharing raw data.

Why this algorithm fits our problem

OUR PROBLEM	HOW THE ALGORITHM HELPS
Data is split across owners	FedAvg trains one shared model without pooling the raw data.
Traffic is a time series	The GRU reads the last hour in order and predicts the next 15 minutes.
Bandwidth is limited	The GRU is small, so each update is only about 14 kB (our estimate).
Sites see different traffic	FedAvg weights clients by data size, and we test uneven data too.

How it solves the problem: one round

1 Send
Server sends the model to all clients.

2 Train
Each client trains on its own data.

3 Return
Clients send back the model, not the data.

4 Combine
Server averages them into a better model.

TINY EXAMPLE: ONE NUMBER INSIDE THE MODEL

Client A
300 samples, learns 0.8

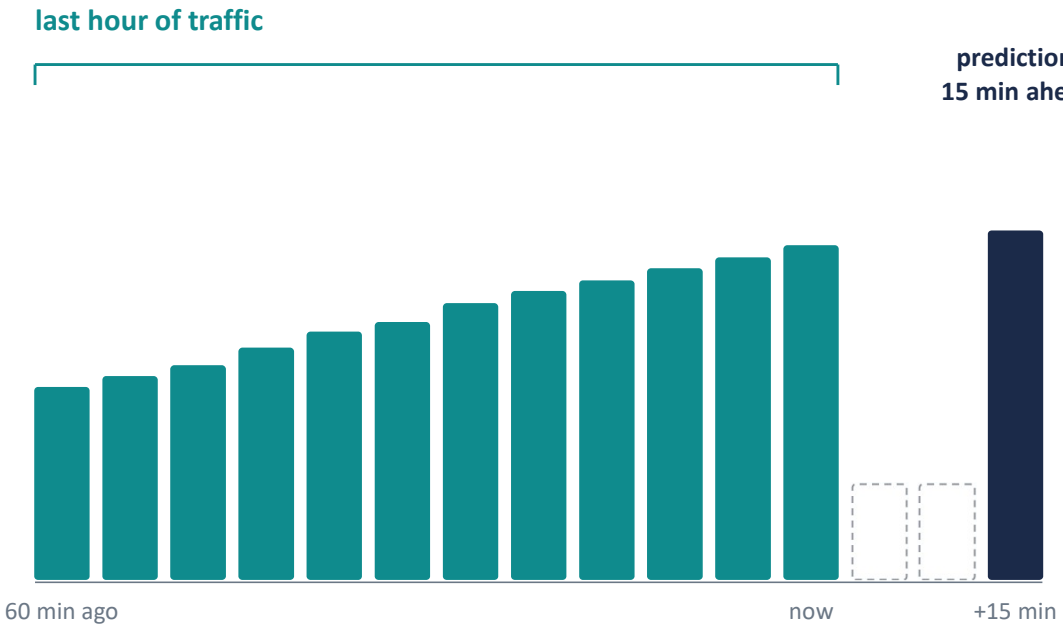
Client B
100 samples, learns 0.4

$$\text{new value} = 0.75 \times 0.8 + 0.25 \times 0.4 = 0.7$$

Client A has 3 times more data, so it counts 3 times more. The real model does this for all its numbers, then repeats for many rounds.

The model: look at the last hour, predict what's next

EXAMPLE



ABOUT GRU

A small neural network that reads data in order, like reading a sentence, and remembers what matters.

Why GRU: it fits time series, and it is small, so each update stays light.

Scored by MAE and RMSE: the average size of the prediction error.

07 WHAT'S DIFFERENT

What makes our approach different

Existing system

- Judged on accuracy only
- Data pooled in one place
- One kind of data assumed
- A single run

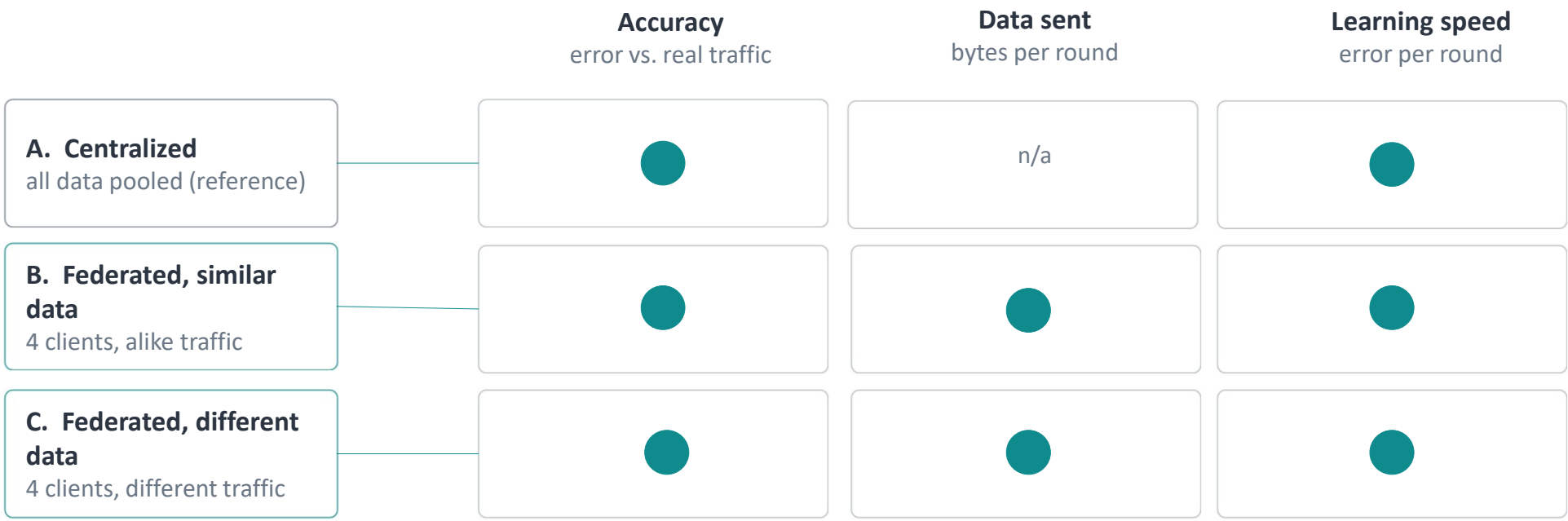
OUR PROJECT

- **Accuracy and data sent, measured together**
- **Data stays on each client**
- **Similar and different client data compared**
- **Same setup for all, repeated 3 times**

Our contribution is the evidence, not a new algorithm.

We do not claim federated learning beats centralized training. We measure the trade-off honestly.

How we will prove it works



Fair test: same dataset (PeMS04) and model. Train on earlier days, test on later days. Each setup runs 3 times.

09 PLAN

What we will build

1 **Prepare data**
load and clean PeMS04

2 **Baseline**
train the normal model

3 **Federate**
add 4 clients and averaging

4 **Test**
run the 3 setups

5 **Dashboard**
show the results

WE WILL DELIVER

Working code, results for the 3 setups, and a simple dashboard.

TOOLS

Python, pandas, PyTorch and Streamlit.

Honest limits. Clients are simulated on one computer, we use one public dataset, and nothing is deployed.

We train one traffic predictor together, without sharing raw data, and measure exactly what it costs.

Prediction Performance

How accurate are the forecasts?

Communication Efficiency

How much data is sent?

Model Convergence

How fast does it learn?

“Smarter Traffic Prediction Through Collaborative Intelligence.”

11 PROTOTYPE

